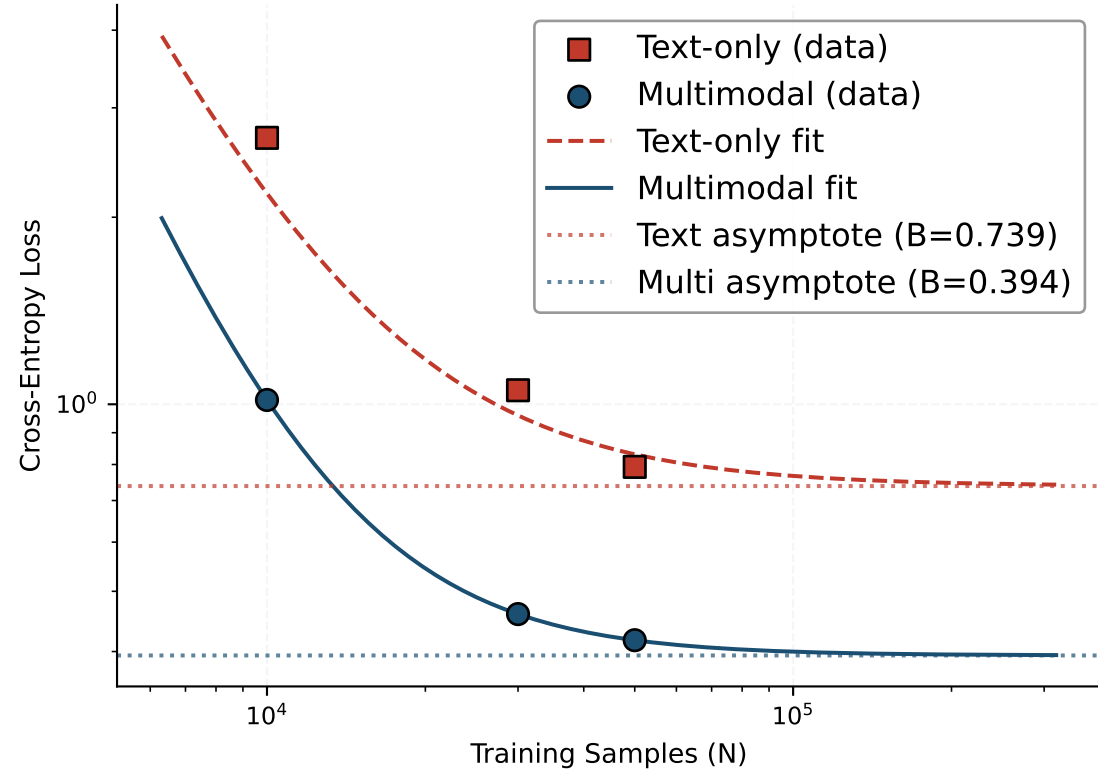
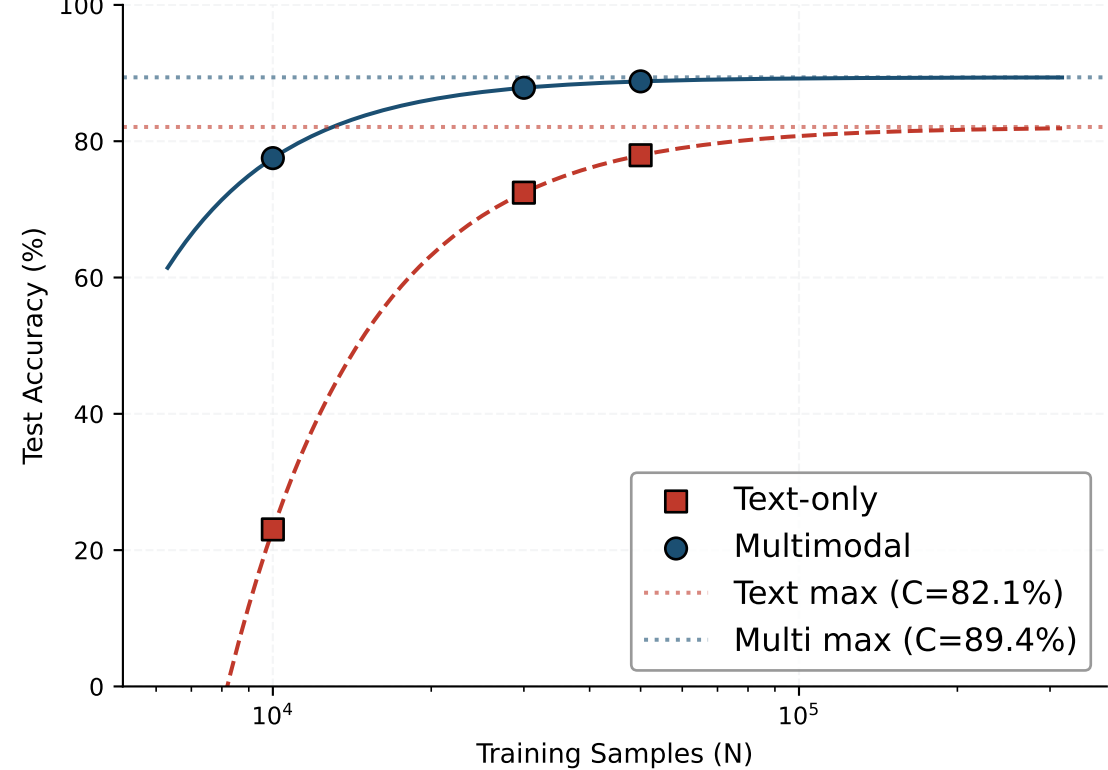


# Scaling Law Analysis: Multimodal vs. Text-Only Document Classification

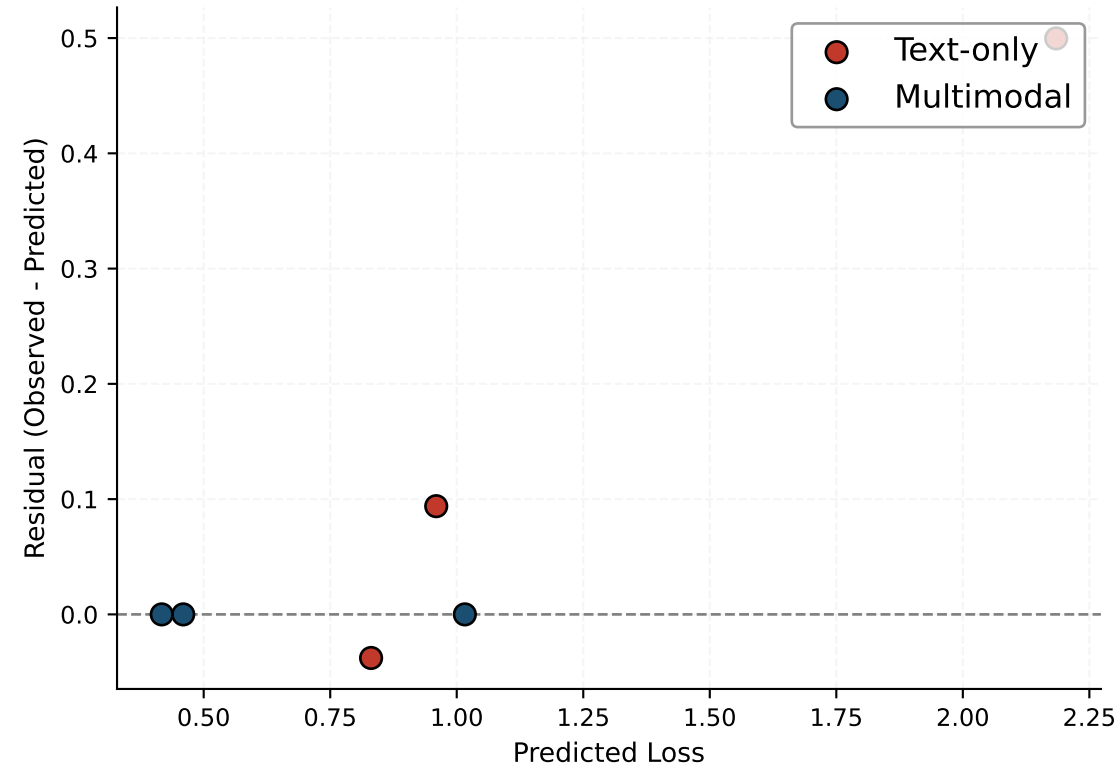
(a) Loss Scaling:  $L(N) = A \cdot N^{(-\alpha)} + B$



(b) Accuracy Scaling:  $ACC(N) = C - D \cdot N^{(-\beta)}$



(c) Residual Analysis



(d) Data Efficiency: Samples to Reach Target

